# LAMMPS official dem10-freefall

Ballistic trajectory and gravity integration | 29 October 2026

| | |
|---|---|
| PASS | Four main runs evaluated using the unchanged official selected-time analytic criteria; source tag unvetted. |

## Official quantities, errors and acceptance limits

| Quantity | Reference / limit | CPU dt | GPU dt | CPU dt/2 | GPU dt/2 |
|---|---|---|---|---|---|
| Freefall z (m) | 0.018608<br>relative limit 1e-09 | 0.018608<br>err 2.174e-13<br>PASS | 0.018608<br>err 2.174e-13<br>PASS | 0.018608<br>err 6.321e-14<br>PASS | 0.018608<br>err 6.246e-14<br>PASS |
| Freefall vz (m/s) | -0.7848<br>relative limit 1e-09 | -0.7848<br>err 1.152e-13<br>PASS | -0.7848<br>err 1.152e-13<br>PASS | -0.7848<br>err 6.024e-13<br>PASS | -0.7848<br>err 6.024e-13<br>PASS |

Check time=0.08 s; original step 8000. Half dt uses twice the check step at the same time. Each cell shows the native value and its own absolute/relative error; a reference limit is not a same-code epsilon.

## Original bodies and source-to-native mapping

| Body | Radius / density | Initial xyz (m) | Initial v (m/s) | Initial spin (rad/s) |
|---|---|---|---|---|
| sphere | R=0.0025<br>rho=2700 | (0, 0, 0.05) | (0, 0, 0) | (0, 0, 0) |

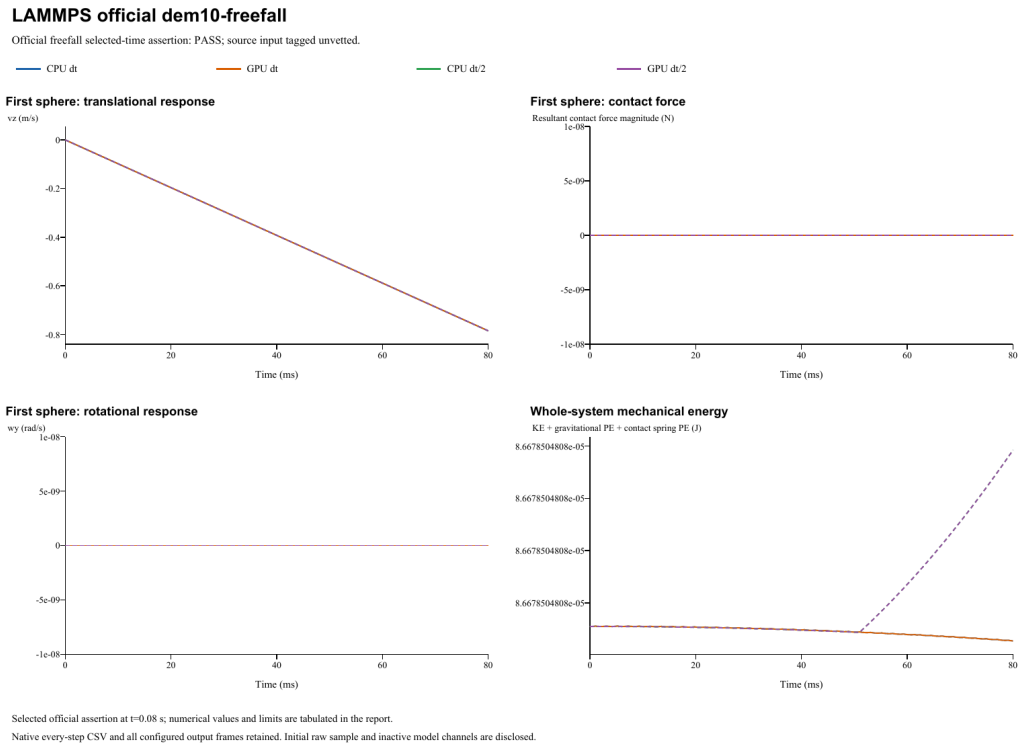
Linear Hooke Fn_elastic=kn*delta, kn=10000 N/m. Automatic damping uses the harmonic material restitution and the active normal/tangential stiffness; stored manual dashpots are inactive. Tangential inputs: kt=0 N/m (explicit in linear mode), mu=0. Rolling/twist coefficients: kr=0, mu_r=0, ktwist=0, mu_twist=0.

Gravity=(0, 0, -9.81) m/s^2. Base/half dt=1e-05/5e-06 s; 8000/16000 steps. No wall and no particle pair; configured Hooke material pair is inactive. Official tight ballistic z/vz assertions are independent.

## Pinned LAMMPS source

Official input | Analytic checker | Parameter mapping.
Commit 29dd3fd7642a6fe9e8b4dd99120df4068b221250; input blob be3161683b34bb6cecb5bc8474e8a89bceebfa1b. Official tag: unvetted. LAMMPS was not rerun locally. Physical inputs were not fitted to native results.

# LAMMPS native histories and refinement




## CPU/GPU and half-step comparison

| Quantity | CPU dt error | CPU dt/2 error | Error type / limit |
|---|---|---|---|
| Freefall z (m) | 2.174e-13 | 6.3206e-14 | relative: 1e-09 |
| Freefall vz (m/s) | 1.1515e-13 | 6.0236e-13 | relative: 1e-09 |

CPU/GPU whole-history maximum position / velocity / spin / force differences: 1.39e-17 m / 0 m/s / 0 rad/s / 0 N.

Maximum individual impulse / position-drift / velocity-kick residuals: 5.86e-17 kg m/s / 3.47e-18 m / 1.06e-16 m/s. Gravity is added separately to saved native contact forces.

## Complete results and verified scope

All 3,204 native frames are retained; main frame counts=[801, 801, 801, 801]. Every numerical-step particle/contact CSV, four loadable main projects and raw initial samples remain preserved. Output cadence is recorded separately from integration dt. Energy/history diagnostics do not replace the official acceptance quantities.

Run manifest | Analysis | Official metrics | Independent audit | Full native ZIP | Frame hashes

Source checkpoints: Source checkpoint states are supplemental regression diagnostics. Their differences are reported directly; the YAML same-code epsilon is not imposed as an independent cross-software acceptance limit.

A stable helper executable is statically linked to the current native CPU/CUDA libraries. Historical GUI/library byte identity is not asserted. Scope is the active source physics and the selected official criteria; inactive options remain outside this test.